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Handledare: Hans Nyqvist

Abstract

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```
Warning: package 'dplyr' was built under R version 4.4.3


Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union

Warning: package 'JMbayes2' was built under R version 4.4.3

Loading required package: survival

Warning: package 'survival' was built under R version 4.4.3

Loading required package: nlme

Warning: package 'nlme' was built under R version 4.4.3


Attaching package: 'nlme'

The following object is masked from 'package:dplyr':

    collapse

Loading required package: GLMMadaptive

Warning: package 'GLMMadaptive' was built under R version 4.4.3

Loading required package: splines


Attaching package: 'JMbayes2'
```

```
The following object is masked from 'package:GLMMadaptive':

    mixed_model

The following object is masked from 'package:nlme':

    lme

The following object is masked from 'package:survival':

    coxph

Warning: package 'lme4' was built under R version 4.4.3

Loading required package: Matrix

Warning: package 'Matrix' was built under R version 4.4.3


Attaching package: 'lme4'

The following object is masked from 'package:GLMMadaptive':

    negative.binomial

The following object is masked from 'package:nlme':

    lmList
```

1 Introduction

1.1 Background

Historically, the detection of AD pathophysiology was done by positron emission topography (PET) to detect the presence of amyloid plaques in the brain, or by analysis of cerbrospinal fluid (CSF) to quantify the concentration of biological molecules implicated in AD pathology. Recent advances in research have made it possible to replace these invasive procedures with blood tests.

According to the National Institute on Aging and the Alzheimer's Association (NIA-AA), there are several core biomarkers to be used in the diagnosis and staging of AD (Ghahremani et al., 2022). One such core biomarker is tau protein phosphorylated at the threonine-181 position (pTAU-181), where concentrations are expected to be higher in individuals with AD pathology (Wong et al., 2012). **MORE ON PTAU-181 here...**

Neurofilament light chain (NfL) is instead a marker of neuronal injury not specific to AD (Ghahremani et al.).

The abnormal concentrations of these biomarkers are the result of the neuropathological changes symptomatic of AD.

General background information. Introducing some domain specific information about AD diagnosis and pathology, so that the reader is well prepared.

1.2 Research Context

Assessing the current landscape in terms of the scientific literature. What is known? What has been done Where are there gaps??? (This leads on to the next section, why are we doing this research, how do we hope to answer some remaining question in the AD research, orient this towards the statistical analysis/approaches!)

1.3 Research Aims

What are our research questions?

1. How do longitudinal changes in plasma biomarker levels over time relate to the risk of progression to dementia? (Joint modeling)
2. How do these associations differ between pTau181 and NfL? (Comparison across biomarkers)
3. How does the precision change for the estimated longitudinal trajectories and survival associations to the inclusion of individuals with different numbers of repeated measurements? (Precision analysis)

Why is this research relevant to the existing body of literature? What does it add? Gaps in previous literature which we outline in above ^^.

How will we accomplish this?

2 Statistical Theory and Methods

Introduce the methods we will use.

Why did we choose this method to answer the research question?

2.1 Longitudinal Data

Longitudinal data in this analysis is defined as n repeated measurements of pTAU-181 and NfL on N individuals in the ADNI1, ADNIGO and ADNI2 cohorts. $T_{i,j}$ and $N_{i,j}$ denote the measured concentrations of pTAU-181 and NfL respectively for the ith individual at the jth occasion. Assuming that the expected concentration of the biomarkers varies with both time and individual level covariates, the mean responses have the notation $\mu_{T:\,i,j} = E(T_{i,j})$ and $\mu_{N:\,i,j} = E(N_{i,j})$. Considering the data are made up of repeated responses on the same individuals, the assumption of independence between observations is illogical. Clearly, the baseline level of pTAU-181 for an individual is correlated to the measurements on subsequent occasions.

2.2 Mixed Effects Models

Why???

We have grouped data, specifically longitudinal data where groups of observations are correlated, since we have repeated measurements on the same individuals. We expect data *within* the groups to be more correlated than those *between* the groups. Mixed effects models are not only for longitudinal data, but for any data where there is non-independence between observations.

2.2.1 Linear Mixed Effect Model

Theory about the model... (specifically why we should use it)

- Biomarker process
- Describes the trajectory of biomarkers at the individual level

Linear mixed effects models? What are these - why use them here?

$$y_i(t) = m_i(t) + \epsilon_i(t) \qquad \epsilon_i(t) \sim N(0, \sigma^2)$$

$$m_i(t) = \mathbf{X_i^T}\beta + \mathbf{Z_i}^T\mathbf{b_i} \qquad \mathbf{b_i}(t) \sim N(0, \sigma^2)$$

$y_i(t)$: the observed longitudinal biomarker

$m_i(t)$: the trajectory function which represents the **true unobserved value of the biomarker** at time t for patient i - we assume that we do not observe the true value due to measurement errors or other reasons.

$\mathbf{X_i}^T(t)\beta$: the fixed effects

$\mathbf{Z_i}^T\mathbf{b_i}$:the random effects

The full model becomes:

$$y_i(t) = \mathbf{X_i^T}\beta + \mathbf{Z_i}^T\mathbf{b_i} + \epsilon_i(t) \qquad \mathbf{b_i}(t) \sim N(0, \sigma^2), \quad \epsilon_i(t) \sim N(0, \sigma^2)$$

2.2.2 Generalised Linear Mixed Effects Model

Ref: https://stats.oarc.ucla.edu/other/mult-pkg/introduction-to-generalized-linear-mixed-models/

An extension of GLMs, where the linear predictor contains random effects alongside the fixed effects. This is used to extend the linear mixed model to non-normal data.

Given random effects, $\mathbf{u}$ the dependent variable, $\mathbf{y}$, is distributed according to the exponential family, such that $g(E[\mathbf{y}|\mathbf{u}]) = \mathbf{X} + \mathbf{Zu}$. Some link function, $g(x)$, links the expected value of $\mathbf{y}$ given the random effects to the linear predictor which included both fixed effects and random effects.

Where:

$\mathbf{X}$ = the fixed effects design matrix

= the fixed effects

$\mathbf{Z}$ = the random effects design matrix

$\mathbf{u}$ = the random effects

What do we model as random?

We can model random intercept and keep all other effects fixed. In that case, for i biomarker measurements on j individuals, the random effects design matrix - $\mathbf{Z}$ - is a $i \cdot j$ matrix containing 1's and 0's, where $z_{i,j} = 1$ if the ith biomarker measurements were taken from the jth individuals, and $z_{i,j} = 0$ if the ith biomarker measurements were not taken from the jth individual.

We do not tend to estimate $\mathbf{u}$, but instead assume that $u_i \sim N(0, \mathbf{G})$, where $\mathbf{G}$ is the variance-covariance matrix of the random effects. The fixed effects are estimated, and the random effects are seen as deviations from the fixed effects and have $E(u) = 0$.

For a model with only a random intercept, $\mathbf{G}$ becomes $G = \sigma^2_{intercept}$. If we extend the model to have random slope, we get:

$$G = \begin{bmatrix} \sigma^2_{intercept} & \sigma_{intercept,slope} \\ \sigma_{intercept,slope} & \sigma^2_{slope} \end{bmatrix}$$

Like all variance-covariance matrices, some elements are redundant since they are repeated. Estimating instead some function of $\mathbf{G}$, where $\mathbf{G} = \sigma(\theta)$, means estimating $\frac{p(p+1)}{2}$ elements instead. Assuming (**?????**) the random effects are independent, the variance-covariance matrix for the random effects becomes diagonal:

$$G = \begin{bmatrix} \sigma^2_{intercept} & 0 \\ 0 & \sigma^2_{slope} \end{bmatrix}$$

2.2.3 Assumptions!!

- Random effects $\mathbf{b_i}$ are **time-invariant** - they are assumed to come from a normal distribution, $\mathbf{b_i} \sim N(0, \sigma^2)$, and represent stable, latent characteristics about the individuals disease, such as a generally fast declining disease trajectory but cannot represent other factors that effect disease trajectory over-time, e.g. medication. Any external time-varying process that effects the biomarker trajectory and the hazard cannot be represented by the random effects.

2.3 Proportional Hazard Model

Theory about the model... (specifically why we should use it)

- Time-to-diagnosis process
- Parametric baseline hazard: **flexible parametrics, splines??**
- OR leave undefined -> Cox PH model

Assuming a Cox PH survival sub-model:

$$h(t|M_i, \mathbf{v_i}) = h_0(t) exp\left\{\phi^T \mathbf{v_i} + \alpha m_i(t)\right\}$$

$\mathbf{v_i}$: vector of time-independent co-variates

ϕ : vector of log-hazard ratios

α : the association parameter

$M_i = \{m_i, \quad 0 \leq s \leq t\}$: the **true unobserved biomarker profile** up to time t

2.3.1 Assumptions!!

- What do we assume about the censoring process?
- Proportionality

2.4 Joint Models

2.4.1 Simple Joint Model

Theory about the model... (specifically why we should use it)

In the joint model, the hazard function at time t is directly dependent on the estimate of the biomarker at time t.

The association structure of the joint model:

- How is the longitudinal model linked to the survival model?
- Does this association structure change for different biomarkers?
- What happens to time-to-diagnosis predictions if we assume the standard structure when another underlying association is a better fit?

If we include the current rate of change alongside the current value in the proportional-hazard model:

$$h(t|M_i, \mathbf{v_i}) = h_0(t) exp\left\{\phi^T \mathbf{v_i} + \alpha_1 m_i(t) + \alpha_2 m_i'(t)\right\} \qquad where \qquad m_i'(t) = \frac{d}{dt} m_i(t)$$

α_1 : links the current value of the biomarker - $m_i(t)$

α_2 : links the current gradient/rate of change - $m_i'(t)$

Joint-model options:

- Longitudinal sub-model: model of class lme()

2.4.2 Assumptions!!

- Full conditional independence
  - Random effects $\mathbf{b_i}$ explain all statistical dependence between (1) the longitudinal outcome and the survival outcome and (2) repeated measurements within the longitudinal process.
  - This assumption can fail when: there are time-varying confounders not captured by the random effects (see the longitudinal model for more info about $\mathbf{b_i}$ assumptions), if the measurement error, $e_i(t)$ in the longitudinal model $y_i(t) = m_i(t) + e_i(t)$ is informative, or there are sub-populations (APOE genotype??? Maybe we need to split into populations based on genotype)
- So: the longitudinal outcome is independent of the time-to-event outcome and the repeated measurements in the longitudinal outcome are independent of each other
- $\mathbf{b_i}$ drives the **true (unobserved)** biomarker trajectory and hazard of AD diagnosis: the observed $y_i(t)$ is the observed biomarker state, $m_i(t)$, plus noise. Where $m_i(t)$ is itself determined by $\mathbf{b_i}$, **since the others are fixed ????? **, once we know $\mathbf{b_i}$, the observed biomarker measurement and the survival times are no longer dependent. The observed y_i's are useful in determining $\mathbf{b_i}$ which is the useful information in estimating survival times.

2.4.3 Multivariate Joint Model

...

3 Data

The data used in this analysis comes from the ADNI cohort under the ADNI1 and ADNI2 protocols. Measurements of the concentration of pTAU-181 and NfL in blood plasma of $n = xxx$ individuals were analysed. In accordance with the ADNI leadership guidelines, which states that there were serious data quality concerns for individuals with a PTID in the format "381_S_10###" , checks were run to assure that no observations with these PTIDs were in the data set (ADNI, 2026).

3.1 ADNI

All protocols allow enrollment for individuals aged 55-90.

- ADNI1

800 subjects (200 controls, 400 MCI, 200 mild AD) recruited at around 50 sites in the United States and Canada

- ADNI2

Cognitively normal, significant memory concern, early MCI, late MCI, mild AD

Inclusion criteria for rollover participants: originally diagnosed as either MCI (early or late) or CN

- ADNI3

700 - 800 rollover participants

370 - 1200 newly enrolled

Three cohorts: CN, MCI, mild AD

- ADNI4

Introducing the study, it's aims and the type of information gathered.

IMPORTANT: REMEMBER WE NEED TO REFERENCE ADNI PROPERLY.

Explain the different protocols and what they mean for data analysis, if we can compare across protocols etc.

We checked whether there were any individuals with a PTID in the format "381_S_10###" as per the ADNI leadership guidelines, which state that due to serious data quality concerns, data from these observations should not be used in any analysis. (SOURCE?)

3.2 Introducing variables... biomarkers

Information about the variables we will use and why, what they can show, how are they recorded

4 Pre-processing

ADNI documentation: https://adni.loni.usc.edu/quick-start-guide-asset/anatomy2.html

Some bullet points to add to the final paper about things we thought about/did:

- Merging the diagnosis data with biomarker/amyloid data by RID (unique identifier for patients)
- Selecting relevant variables from diagnosis and biomarker/amyloid data set
- What variables do we need for longitudinal sub-model? (Repeated measurement, time followed?)
- What variables do we need for survival sub-model? (Event, time followed?)
- Joint model: time-var: a character string indicating the time variable in the mixed-effects model
- How many observations going from CN/MCI -> AD? Do we need a multi-state survival model?

5 Results

Present the results from our analyses. Do not interpret results here! That should be done in the discussion.

6 Discussion

6.1 Main findings/summary

Interpret the results. What did we find?

6.2 Discussion - Relating to the research question(s)/previous literature

How do our results help to answer the research question? Anything unexpected/not in accordance with previous studies?

6.3 Further research

Suggest what kinds of questions this analysis leaves us with? What more can be done/looked at?

7 Conclusions

Summary

8 Bibliography

Ghahremani M, Wang M, Chen HY, Zetterberg H, Smith E, Ismail Z; Alzheimer's Disease Neuroimaging Initiative*. Plasma Phosphorylated Tau at Threonine 181 and Neuropsychiatric Symptoms in Preclinical and Prodromal Alzheimer Disease. *Neurology.* 2023 Feb 14;100(7):e683-e693. doi: 10.1212/WNL.0000000000201517.

Wong, P.C., Savonenko, A., Li, T., Price, DL. (2012). 'Neurobiology of Alzheimer's Disease' in Brady, S.T., Siegel, G.J., Albers, RW., Price, D.L (8 ed.) *Basic neurochemistry : principles of molecular, cellular, and medical neurobiology.* Waltham, Massachusetts; Oxford: Academic Press / Elsevier.

9 Pre-processing

9.1 Checking RID with more than one measurement recorded on the same exam date

- There are 52 individuals with two measurements recorded on the same exam date.
- How should we deal with these? Average the measurements? Remove the observations?
- Some of the values are quite different, what do we assume by averaging them? Do we introduce bias but just just removing them?
- For now, remove them:

```
integer(0)
```

9.2 We select only the required variables and merge them

```
Warning in left_join(., ptau_data, by = c("RID", "VISCODE2"), suffix = c("_nfl", : Detected a
to-many relationship between `x` and `y`.
i Row 738 of `x` matches multiple rows in `y`.
i Row 738 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

```
Warning in left_join(., diagnosis_data_dx, by = c("RID", "VISCODE2")): Detected an unexpecte
to-many relationship between `x` and `y`.
i Row 554 of `x` matches multiple rows in `y`.
i Row 5172 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

9.3 Concerning 381_S_10

9.4 We check the correlation between the biomarkers

```
[1] 0.2472172
```

9.5 We add selected demographics variables (age + gender), to later use in survival model and longitudinal model

9.6 We filter participants with at least 3 measurements

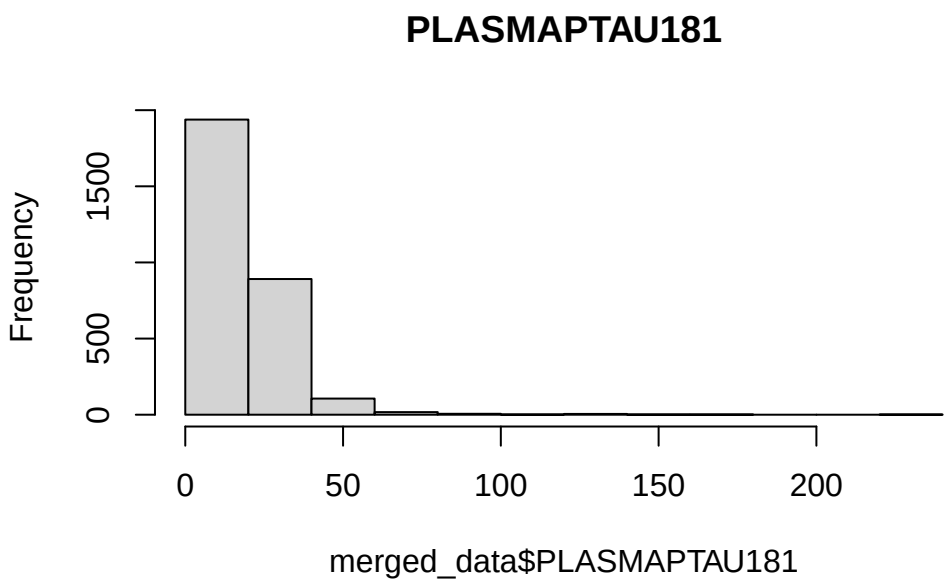
```
[1] 768
```

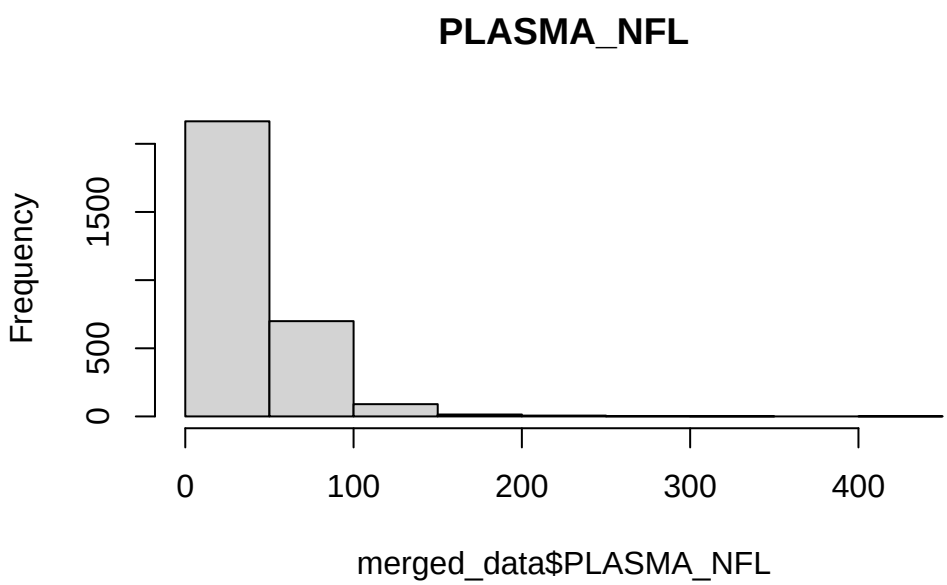
9.7 We extract baseline visits, and create an event variable and a time-to-event variable

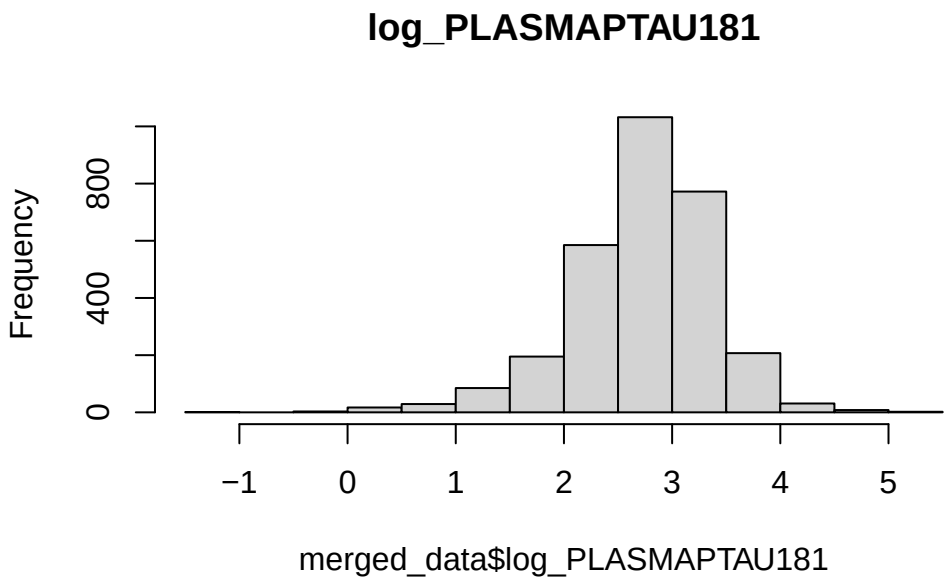
```
Warning: There were 577 warnings in `summarise()`.
The first warning was:
i In argument: `time_to_event = `/`(...)`.
i In group 1: `RID = 2`.
Caused by warning in `min.default()`:
! no non-missing arguments to min; returning Inf
i Run `dplyr::last_dplyr_warnings()` to see the 576 remaining warnings.


  0   1
577 178
```

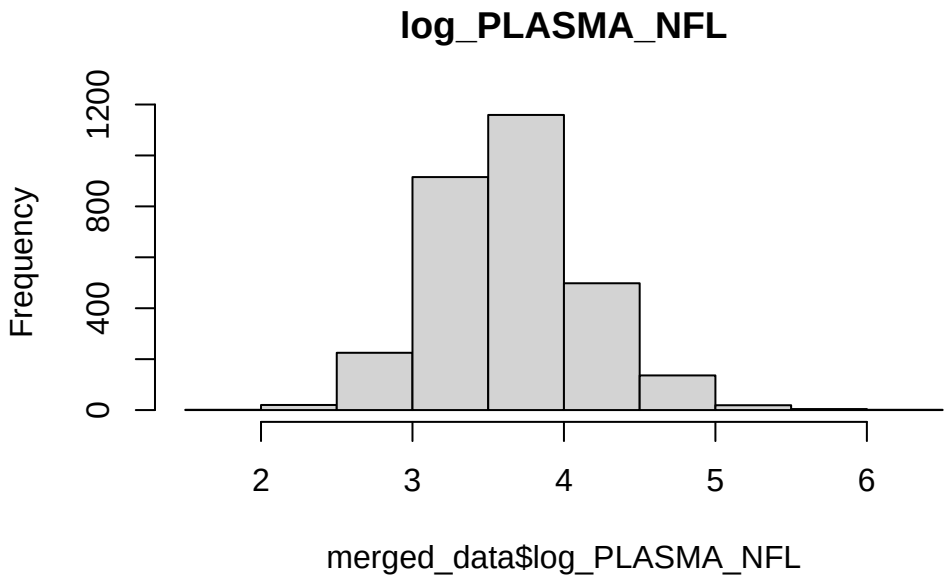
9.8 We create a time variable (time from baseline)

9.9 We check the distribution of each relevant variable for multivariate joint model and transform









9.10 We check for outliers

We still keep them, as we assume they are correct observations.

```
# A tibble: 46 x 25
     RID VISCODE2 EXAMDATE_nfl PLASMA_NFL EXAMDATE_ptau PLASMAPTAU181
   <dbl> <chr>    <date>            <dbl> <date>                <dbl>
 1   125 m48      2010-03-02         46.2 2010-03-02             76.1
 2   159 m108     2015-03-05         48.2 2015-03-05            238.
 3   159 m60      2011-04-28         36.2 2011-04-28            124.
 4   159 m72      2012-03-05         36.5 2012-03-05            137.
 5   159 m96      2014-03-12         38.9 2014-03-12            169.
 6   605 m108     2015-06-30         65.3 2015-06-30            110.
 7   605 m84      2013-07-16         53.5 2013-07-16            139.
 8   658 m72      2012-08-28         35.9 2012-08-28             69.5
 9   658 m96      2014-10-03         46.7 2014-10-03             62.9
10   709 m72      2012-08-09         59.3 2012-08-09             86.7
# i 36 more rows
# i 19 more variables: ORIGPROT_dx <chr>, COLPROT_dx <fct>, EXAMDATE_dx <date>,
#   DIAGNOSIS_dx <chr>, PTDOBYY <dbl>, PTGENDER.x <fct>, age_at_visit <dbl>,
#   APOE4_status.x <dbl>, baseline_date <date>, age_at_baseline <dbl>,
#   PTGENDER.y <fct>, APOE4_status.y <dbl>, time_to_event <dbl>, event <int>,
#   time_years <dbl>, log_PLASMAPTAU181 <dbl>, log_PLASMA_NFL <dbl>,
#   ptau_z <dbl[,1]>, nfl_z <dbl[,1]>
```

```
# A tibble: 44 x 25
     RID VISCODE2 EXAMDATE_nfl PLASMA_NFL EXAMDATE_ptau PLASMAPTAU181
   <dbl> <chr>    <date>            <dbl> <date>                <dbl>
 1     8 m60      2010-09-27         196. 2010-09-27             18.3
 2   126 m108     2015-03-27         128. 2015-03-27             27.2
 3   126 m60      2011-03-17         134. 2011-03-17             26.4
 4   126 m84      2013-04-04         203. 2013-04-04             30.0
 5   230 m108     2015-02-03         142. 2015-02-03             37.4
 6   301 m96      2014-04-14         227. 2014-04-14             13.8
 7   307 m72      2012-06-21         246. 2012-06-21             37.1
 8   307 m84      2013-07-02         258. 2013-07-02             48.3
 9   307 m96      2014-06-24         306. 2014-06-24             48.8
10   558 m72      2012-08-13         144. 2012-08-13             30.1
# i 34 more rows
# i 19 more variables: ORIGPROT_dx <chr>, COLPROT_dx <fct>, EXAMDATE_dx <date>,
#   DIAGNOSIS_dx <chr>, PTDOBYY <dbl>, PTGENDER.x <fct>, age_at_visit <dbl>,
#   APOE4_status.x <dbl>, baseline_date <date>, age_at_baseline <dbl>,
#   PTGENDER.y <fct>, APOE4_status.y <dbl>, time_to_event <dbl>, event <int>,
#   time_years <dbl>, log_PLASMAPTAU181 <dbl>, log_PLASMA_NFL <dbl>,
#   ptau_z <dbl[,1]>, nfl_z <dbl[,1]>
```

9.11 We check for missing values in both merged_data and long_data

```
            RID          VISCODE2      EXAMDATE_nfl        PLASMA_NFL
              0                 0                 0                 0
   EXAMDATE_ptau     PLASMAPTAU181       ORIGPROT_dx        COLPROT_dx
             11                11                18                18
     EXAMDATE_dx      DIAGNOSIS_dx           PTDOBYY        PTGENDER.x
             18                18                 0                 0
    age_at_visit    APOE4_status.x     baseline_date   age_at_baseline
             11                 0                42                 0
      PTGENDER.y    APOE4_status.y     time_to_event             event
              0                 0                94                51
      time_years log_PLASMAPTAU181    log_PLASMA_NFL            ptau_z
             42                11                 0                11
           nfl_z
               0

          RID age_at_baseline        PTGENDER    APOE4_status   time_to_event
            0               0               0               0              24
        event
           13

            RID          VISCODE2      EXAMDATE_nfl        PLASMA_NFL
              0                 0                 0                 0
   EXAMDATE_ptau     PLASMAPTAU181       ORIGPROT_dx        COLPROT_dx
              0                 0                 0                 0
     EXAMDATE_dx      DIAGNOSIS_dx           PTDOBYY        PTGENDER.x
              0                 0                 0                 0
    age_at_visit    APOE4_status.x     baseline_date   age_at_baseline
              0                 0                 0                 0
      PTGENDER.y    APOE4_status.y     time_to_event             event
              0                 0                 0                 0
      time_years log_PLASMAPTAU181    log_PLASMA_NFL            ptau_z
              0                 0                 0                 0
           nfl_z
               0

          RID age_at_baseline        PTGENDER    APOE4_status   time_to_event
            0               0               0               0               0
        event
            0
```

9.12 We make sure that merged_data and surv_data have the same length (for the joint model)

```
[1] 744
```

```
[1] 744
```

10 Longitudinal model

10.1 Fitted LME model for P-TAU

```
Call:
GLMMadaptive::mixed_model(fixed = log_PLASMAPTAU181 ~ time_years +
    PTGENDER.y + APOE4_status.y, random = ~time_years | RID,
    data = merged_data, family = students.t(df = 3, link = "identity"))

Data Descriptives:
Number of Observations: 2884
Number of Groups: 744

Model:
 family: Student's-t
 link: identity

Fit statistics:
  log.Lik      AIC      BIC
 -1849.33 3714.659 3751.555

Random effects covariance matrix:
             StdDev    Corr
(Intercept)  0.4867
time_years   0.0274  0.0367

Fixed effects:
               Estimate Std.Err z-value   p-value
(Intercept)      2.5262  0.0332 76.2047   < 1e-04
time_years       0.0319  0.0044  7.1631   < 1e-04
PTGENDER.yMale   0.1144  0.0382  2.9936 0.0027575
APOE4_status.y   0.2787  0.0384  7.2521   < 1e-04
```

```
phi parameters:
      Estimate Std.Err
phi_1  -1.5003  0.0237

Integration:
method: adaptive Gauss-Hermite quadrature rule
quadrature points: 11

Optimization:
method: hybrid EM and quasi-Newton
converged: TRUE
```

10.2 Fitted LME model for NFL

```
Linear mixed-effects model fit by REML
  Data: merged_data
       AIC      BIC    logLik
  1625.768 1673.492 -804.8841

Random effects:
 Formula: ~time_years | RID
 Structure: General positive-definite, Log-Cholesky parametrization
            StdDev     Corr
(Intercept) 0.43113927 (Intr)
time_years  0.03255857 0.248
Residual    0.21806581

Fixed effects:  log_PLASMA_NFL ~ time_years + PTGENDER.y + APOE4_status.y
                  Value  Std.Error   DF   t-value p-value
(Intercept)    3.553066 0.02885685 2139 123.12730  0.0000
time_years     0.044541 0.00350788 2139  12.69747  0.0000
PTGENDER.yMale 0.007528 0.03372888  741   0.22321  0.8234
APOE4_status.y 0.057456 0.03402772  741   1.68851  0.0917
 Correlation:
               (Intr) tm_yrs PTGEND
time_years     -0.116
PTGENDER.yMale -0.641  0.001
APOE4_status.y -0.478 -0.001 -0.017

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-4.33485366 -0.46792878 -0.04377811  0.40022783  7.85889764
```

```
Number of Observations: 2884
Number of Groups: 744
```

10.3 Removing gender from NFL LME model

We see that gender is not significant in the LME model for NFL, so we remove it:

```
Linear mixed-effects model fit by REML
  Data: merged_data
       AIC      BIC    logLik
  1618.876 1660.638 -802.4382

Random effects:
 Formula: ~time_years | RID
 Structure: General positive-definite, Log-Cholesky parametrization
            StdDev     Corr
(Intercept) 0.43084249 (Intr)
time_years  0.03255538 0.248
Residual    0.21806687

Fixed effects:  log_PLASMA_NFL ~ time_years + APOE4_status.y
                  Value  Std.Error   DF   t-value p-value
(Intercept)    3.557193 0.02213448 2139 160.70826  0.0000
time_years     0.044540 0.00350784 2139  12.69736  0.0000
APOE4_status.y 0.057585 0.03400119  742   1.69363  0.0908
 Correlation:
               (Intr) tm_yrs
time_years     -0.151
APOE4_status.y -0.636 -0.001

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-4.33444238 -0.46757581 -0.04447896  0.39928847  7.85719091

Number of Observations: 2884
Number of Groups: 744
```

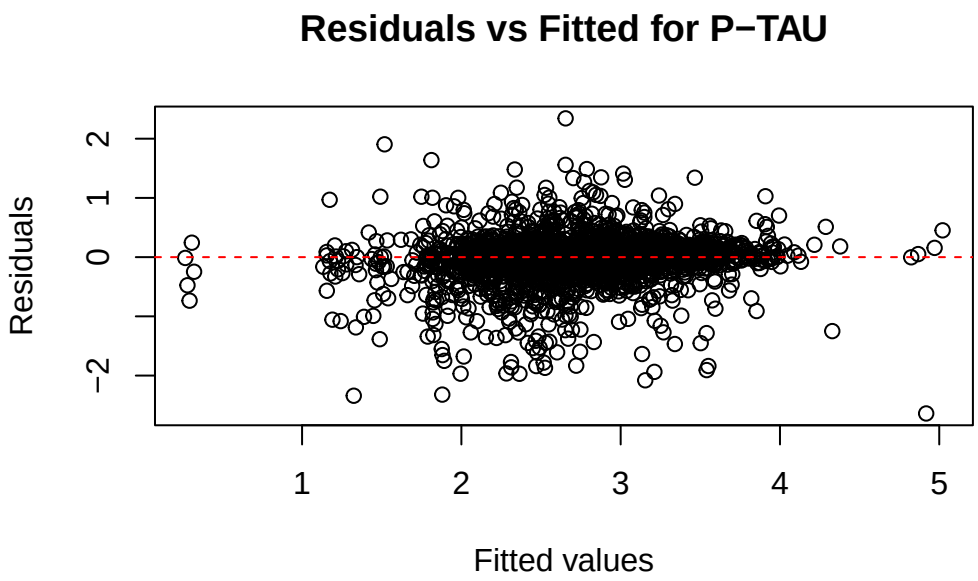
10.4 Removing APOE4 status from ptau LME model

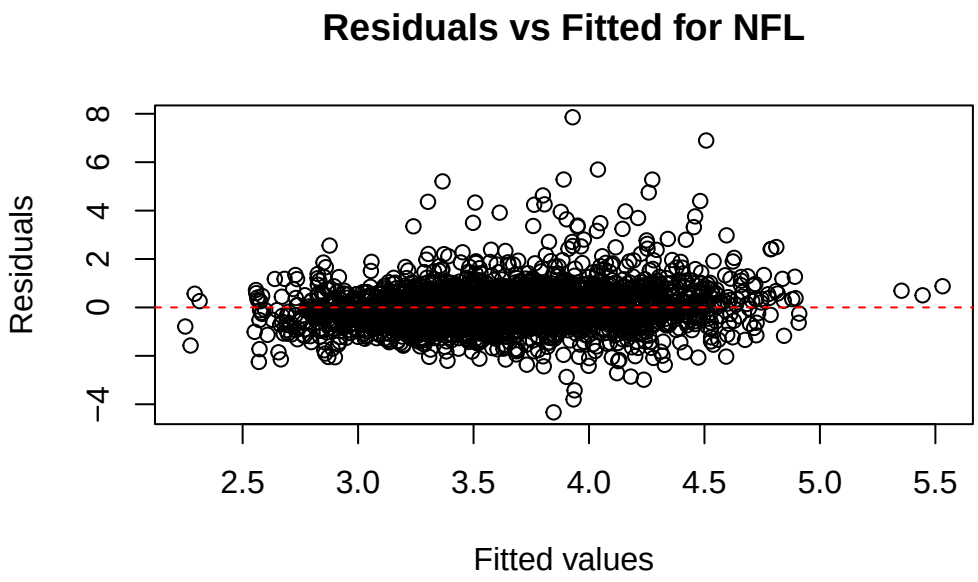
Even though APOE4 status was significant in the LME model for ptau, including it in both the LME model and the survival model led to poor MCMC convergence (high Rhat) in the joint model, probably due to difficulties in separating its direct effect on survival from its indirect effect through the biomarker. Therefore, to ensure stable estimation, we keep the APOE4 status only in the survival model, removing it from the LME modell:

```
Call:
GLMMadaptive::mixed_model(fixed = log_PLASMAPTAU181 ~ time_years +
    PTGENDER.y, random = ~time_years | RID, data = merged_data,
    family = students.t(df = 3, link = "identity"))

Data Descriptives:
Number of Observations: 2884
Number of Groups: 744

Model:
 family: Student's-t
 link: identity

Fit statistics:
   log.Lik      AIC      BIC
 -1874.616 3763.231 3795.516

Random effects covariance matrix:
             StdDev    Corr
(Intercept)  0.5050
time_years   0.0166  0.1626

Fixed effects:
               Estimate Std.Err z-value   p-value
(Intercept)      2.6408  0.0302 87.5418   < 1e-04
time_years       0.0320  0.0044  7.3338   < 1e-04
PTGENDER.yMale   0.1175  0.0396  2.9684 0.0029933

phi parameters:
      Estimate Std.Err
phi_1  -1.4938  0.0234

Integration:
```

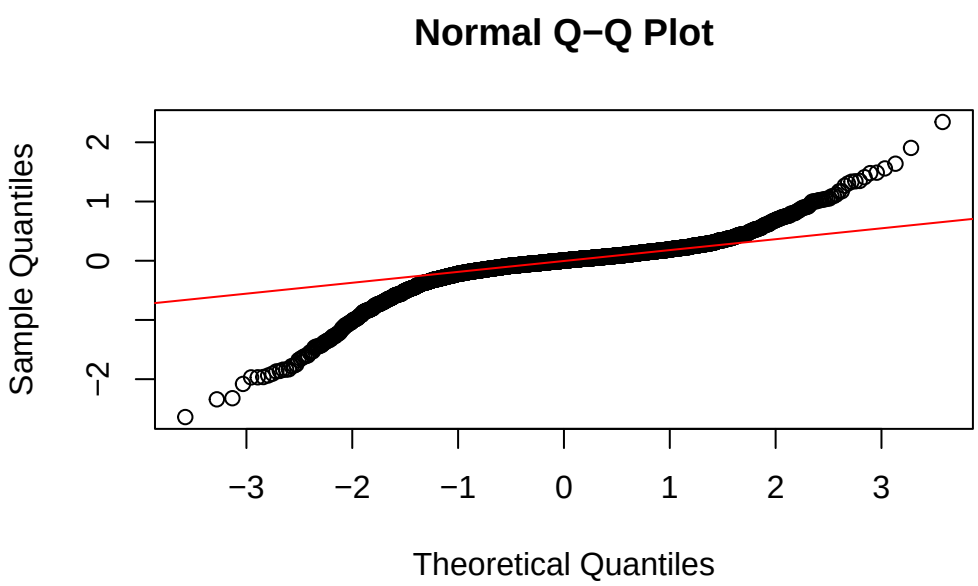
```
method: adaptive Gauss-Hermite quadrature rule
quadrature points: 11

Optimization:
method: hybrid EM and quasi-Newton
converged: TRUE
```

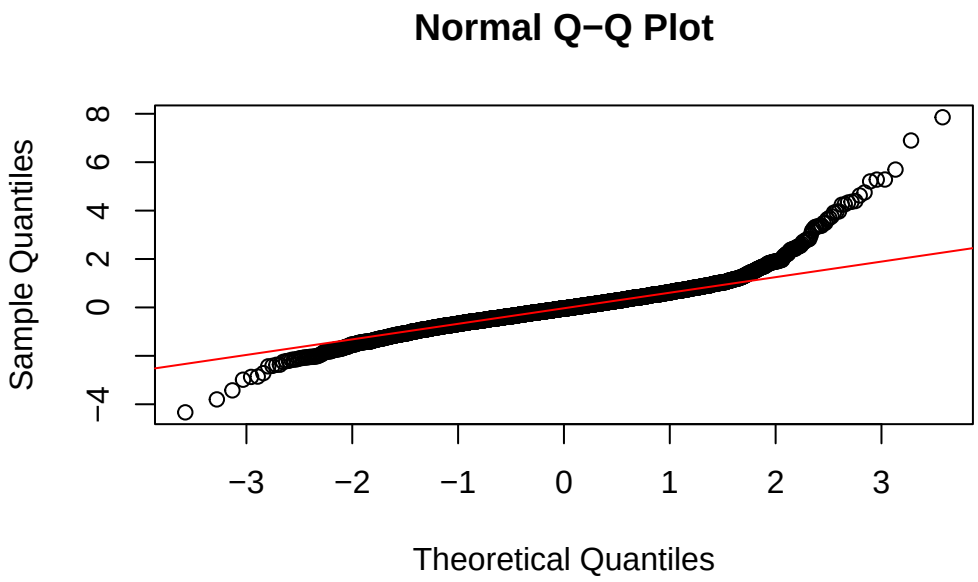
10.5 Residuals




Residuals vs Fitted for NFL
Residuals
8
6
4
2
0
−4
2.5
3.0
3.5
4.0
4.5
5.0
5.5
Fitted values

10.6 6. QQ plot

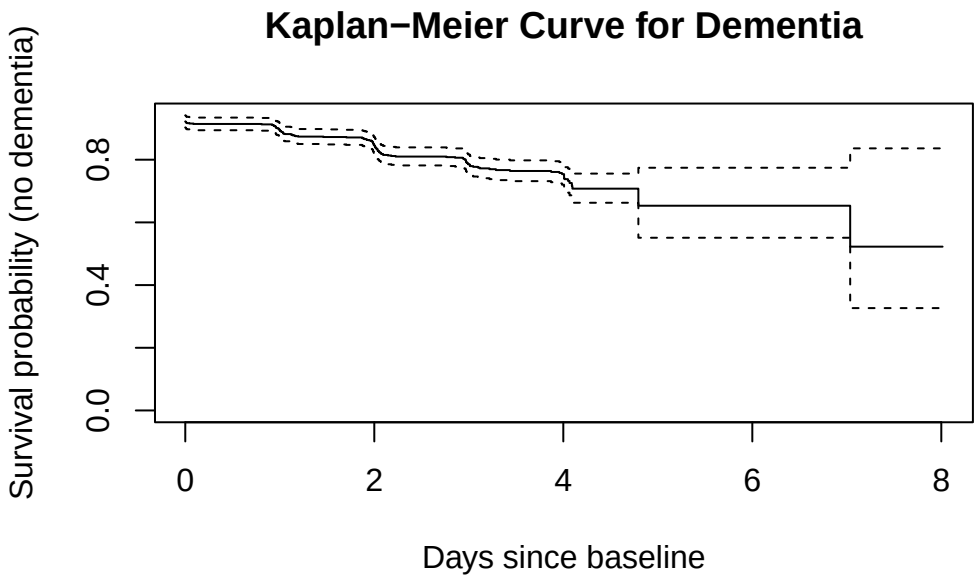

Normal Q−Q Plot
Sample Quantiles
2
1
0
−2
−3
−2
−1
0
1
2
3
Theoretical Quantiles


Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−4
0
2
4
6
8
−3
−2
−1
0
1
2
3

11 Survival model

11.1 We create a survival object

11.2 Kaplan-Meier curve



11.3 Fitted Coxph model

```
Call:
survival::coxph(formula = surv_object ~ age_at_baseline + PTGENDER +
    APOE4_status, data = data <- surv_data)

  n= 744, number of events= 172

                   coef exp(coef) se(coef)     z Pr(>|z|)
age_at_baseline 0.03902   1.03979  0.01059 3.683  0.00023 ***
PTGENDERMale    0.04294   1.04387  0.15747 0.273  0.78510
APOE4_status    1.37877   3.97002  0.16583 8.314  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                exp(coef) exp(-coef) lower .95 upper .95
```

```
age_at_baseline     1.040     0.9617    1.0184     1.062
PTGENDERMale        1.044     0.9580    0.7667     1.421
APOE4_status        3.970     0.2519    2.8684     5.495

Concordance= 0.711  (se = 0.02 )
Likelihood ratio test= 82.56  on 3 df,   p=<2e-16
Wald test            = 76.51  on 3 df,   p=<2e-16
Score (logrank) test = 85.72  on 3 df,   p=<2e-16
```

11.4 Removing gender

Gender is not significant in the survival model, so we remove it:

```
Call:
survival::coxph(formula = surv_object ~ age_at_baseline + APOE4_status,
    data = data <- surv_data)

  n= 744, number of events= 172

                   coef exp(coef) se(coef)     z Pr(>|z|)
age_at_baseline 0.03947   1.04026  0.01047 3.769 0.000164 ***
APOE4_status    1.38169   3.98164  0.16553 8.347  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                exp(coef) exp(-coef) lower .95 upper .95
age_at_baseline     1.040     0.9613     1.019     1.062
APOE4_status        3.982     0.2512     2.878     5.508

Concordance= 0.71  (se = 0.02 )
Likelihood ratio test= 82.48  on 2 df,   p=<2e-16
Wald test            = 76.35  on 2 df,   p=<2e-16
Score (logrank) test = 85.54  on 2 df,   p=<2e-16
```

11.5 Testing for the assumption of proportional hazards

```
                chisq df     p
age_at_baseline 0.254  1 0.614
APOE4_status    3.930  1 0.047
GLOBAL          4.572  2 0.102
```

12 Joint-model

12.1 Fitted joint model for P-tau

```
Call:
JMbayes2::jm(Surv_object = surv_model, Mixed_objects = long_model_ptau,
    time_var = "time_years", n_chains = 3L, n_iter = 11000L,
    n_burnin = 1000L)

Data Descriptives:
Number of Groups: 744       Number of events: 172 (23.1%)
Number of Observations:
  log_PLASMAPTAU181: 2884

                  DIC      WAIC       LPML
marginal    4048.8616 4434.1350 -2283.8492
conditional  345.1709  384.1125  -905.2406

Random-effects covariance matrix:

       StdDev   Corr
(Intr) 0.5103 (Intr)
tm_yrs 0.0146 -0.1671

Survival Outcome:
                           Mean  StdDev    2.5%  97.5%      P   Rhat
age_at_baseline          0.0159 0.0182 -0.0193 0.0509 0.3983 1.0033
APOE4_status             2.2694 0.3054  1.6978 2.8981 0.0000 1.0220
value(log_PLASMAPTAU181) 3.3170 0.5802  2.3744 4.5827 0.0000 1.0555

Longitudinal Outcome: log_PLASMAPTAU181 (family = Student's-t, link = identity)
                  Mean  StdDev   2.5%  97.5%      P   Rhat
(Intercept)    2.6533 0.0304 2.5934 2.7127 0.0000 1.0161
time_years     0.0279 0.0050 0.0172 0.0365 0.0000 1.4018
PTGENDER.yMale 0.1172 0.0391 0.0409 0.1945 0.0027 1.0012

MCMC summary:
chains: 3
iterations per chain: 11000
burn-in per chain: 1000
thinning: 1
```

```
time: 2.9 min
```

12.2 Fitted joint model for NFL

```
Call:
JMbayes2::jm(Surv_object = surv_model, Mixed_objects = long_model_nfl,
    time_var = "time_years", n_chains = 3L, n_iter = 11000L,
    n_burnin = 1000L)

Data Descriptives:
Number of Groups: 744       Number of events: 172 (23.1%)
Number of Observations:
  log_PLASMA_NFL: 2884

                  DIC      WAIC        LPML
marginal     2098.859  2301.524 -1188.25641
conditional -1179.452 -1363.147    51.96947

Random-effects covariance matrix:

       StdDev   Corr
(Intr) 0.4340 (Intr)
tm_yrs 0.0274 0.2577

Survival Outcome:
                         Mean  StdDev    2.5%  97.5%      P   Rhat
age_at_baseline       -0.0175 0.0193 -0.0547 0.0204 0.3735 1.0052
APOE4_status           2.1881 0.2567  1.6966 2.6945 0.0000 1.0094
value(log_PLASMA_NFL)  2.5701 0.3040  1.9901 3.2005 0.0000 1.0134

Longitudinal Outcome: log_PLASMA_NFL (family = gaussian, link = identity)
                 Mean  StdDev    2.5%  97.5%      P   Rhat
(Intercept)    3.5613 0.0223  3.5178 3.6050 0.0000 1.0040
time_years     0.0426 0.0036  0.0356 0.0495 0.0000 1.0392
APOE4_status.y 0.0614 0.0340 -0.0050 0.1280 0.0707 1.0004
sigma          0.2199 0.0039  0.2122 0.2275 0.0000 1.0847

MCMC summary:
chains: 3
iterations per chain: 11000
burn-in per chain: 1000
```

```
thinning: 1
time: 2.4 min
```

13 Multivariate joint model

13.1 We create a mixed object

13.2 We check wether the models have stable random slopes

```
Call:
GLMMadaptive::mixed_model(fixed = log_PLASMAPTAU181 ~ time_years +
    PTGENDER.y, random = ~time_years | RID, data = merged_data,
    family = students.t(df = 3, link = "identity"))

Data Descriptives:
Number of Observations: 2884
Number of Groups: 744

Model:
 family: Student's-t
 link: identity

Fit statistics:
   log.Lik      AIC      BIC
 -1874.616 3763.231 3795.516

Random effects covariance matrix:
             StdDev    Corr
(Intercept)  0.5050
time_years   0.0166  0.1626

Fixed effects:
               Estimate Std.Err z-value   p-value
(Intercept)      2.6408  0.0302 87.5418   < 1e-04
time_years       0.0320  0.0044  7.3338   < 1e-04
PTGENDER.yMale   0.1175  0.0396  2.9684 0.0029933

phi parameters:
      Estimate Std.Err
phi_1  -1.4938  0.0234
```

```
Integration:
method: adaptive Gauss-Hermite quadrature rule
quadrature points: 11

Optimization:
method: hybrid EM and quasi-Newton
converged: TRUE


RID = pdLogChol(time_years)
            Variance    StdDev     Corr
(Intercept) 0.185625252 0.43084249 (Intr)
time_years  0.001059853 0.03255538 0.248
Residual    0.047553158 0.21806687
```

For both models, the slope variance is very low (lme: 0.0011, mixed_model: 0.036), which indicates that individual biomarker trajectories over time show minimal variation –> We skip the random slope in fForms:

13.3 fForms

13.4 Fitted multivariate joint model

```
Call:
JMbayes2::jm(Surv_object = surv_model, Mixed_objects = Mixed,
    time_var = "time_years", functional_forms = fForms, n_chains = 3L,
    n_iter = 20000L, n_burnin = 2000L)

Data Descriptives:
Number of Groups: 744       Number of events: 172 (23.1%)
Number of Observations:
  log_PLASMAPTAU181: 2884
  log_PLASMA_NFL: 2884

                  DIC      WAIC        LPML
marginal     5416.150 10586.910 -6419.00758
conditional -2517.146 -2201.048   -46.79348

Random-effects covariance matrix:
```

```
       StdDev   Corr
(Intr) 0.5086 (Intr)  tm_yrs  (Intr)
tm_yrs 0.0164 -0.0179
(Intr) 0.4365 0.4779  -0.0647
tm_yrs 0.0297 0.3003  0.4499  0.2396

Survival Outcome:
                           Mean  StDev    2.5%  97.5%      P   Rhat
age_at_baseline          -0.0223 0.0202 -0.0623 0.0164 0.2708 1.0000
APOE4_status              2.1355 0.2881  1.5961 2.7340 0.0000 1.0082
value(log_PLASMAPTAU181)  2.3973 0.5057  1.6352 3.4949 0.0000 1.1030
value(log_PLASMA_NFL)     1.5194 0.3553  0.8072 2.2099 0.0001 1.0077

Longitudinal Outcome: log_PLASMAPTAU181 (family = Student's-t, link = identity)
                 Mean  StDev   2.5%  97.5%     P   Rhat
(Intercept)    2.6508 0.0282 2.5958 2.7067 0.000 1.0066
time_years     0.0296 0.0040 0.0215 0.0370 0.000 1.1261
PTGENDER.yMale 0.1158 0.0349 0.0479 0.1840 0.001 1.0007

Longitudinal Outcome: log_PLASMA_NFL (family = gaussian, link = identity)
                  Mean  StDev    2.5%  97.5%      P   Rhat
(Intercept)     3.6038 0.0215  3.5617 3.6458 0.0000 1.0007
time_years      0.0440 0.0034  0.0370 0.0503 0.0000 1.0279
APOE4_status.y -0.0479 0.0315 -0.1100 0.0142 0.1289 1.0000
sigma           0.2189 0.0040  0.2112 0.2270 0.0000 1.2947

MCMC summary:
chains: 3
iterations per chain: 20000
burn-in per chain: 2000
thinning: 1
time: 7.7 min

                               Mean      StDev        2.5%      97.5%
age_at_baseline          -0.0223351 0.02019721 -0.06228694 0.01644353
APOE4_status              2.1355289 0.28808743  1.59605667 2.73401096
value(log_PLASMAPTAU181)  2.3972791 0.50573700  1.63522488 3.49490131
value(log_PLASMA_NFL)     1.5193962 0.35525441  0.80724503 2.20989065
                                   P     Rhat
age_at_baseline          0.2708148148 1.000049
APOE4_status             0.0000000000 1.008243
value(log_PLASMAPTAU181) 0.0000000000 1.102964
value(log_PLASMA_NFL)    0.0001111111 1.007719
```